# CISPLATIN

Highly emetogenic

Nephro-, oto-, neurotoxic

**CHEMICAL NAME** cis-diamminedichloroplatinum(II) — cis-$[PtCl_2(NH_3)_2]$
**CLASS** Platinum-based alkylating-like cytotoxic agent
**TRADE NAMES** Platinol · Cisplatin injection · Platosin

## THERAPEUTIC USES / INDICATIONS

- Testicular cancer (curative), ovarian, cervical (with radiotherapy), bladder, head and neck, lung (NSCLC, SCLC), oesophageal, gastric cancers; osteosarcoma.

## ADMINISTRATION

- **IV** infusion with aggressive pre- and post-hydration (with magnesium/potassium, ± mannitol); maintain urine output >100 mL/h.
- Do not use aluminium needles (precipitate); protect from light.
- Saline diluent (chloride stabilises).
- Cytotoxic precautions.

## DOSAGES

- **Typical:** 50–100 mg/m² IV every 3–4 weeks, or 20 mg/m² daily × 5 days (testicular BEP).
- **Weekly with radiotherapy:** 40 mg/m².
- Contraindicated in pre-existing renal impairment; check renal function before each course.

## PHARMACOKINETICS

- Highly protein-bound; concentrates in kidney, liver.
- Excreted in urine. **t½:** terminal ~5 days (platinum retained for months).

## PHARMACODYNAMICS

- Forms **platinum–DNA cross-links** (intra- and inter-strand) → blocks DNA replication and transcription → apoptosis.

## SIDE & ADVERSE EFFECTS

### SIDE EFFECTS (COMMON)

- Severe nausea and vomiting (acute and delayed).
- Fatigue, anorexia.
- Taste changes, alopecia (mild).

### ADVERSE EFFECTS (SERIOUS)

- Nephrotoxicity (dose-limiting); AKI, hypomagnesaemia, hypokalaemia.
- Ototoxicity (high-frequency hearing loss, tinnitus) — especially children.
- Peripheral neuropathy (cumulative).
- Myelosuppression (moderate); anaphylaxis; secondary leukaemia.

## CONTRAINDICATIONS & CAUTIONS

- Pre-existing renal impairment (CrCl <60), hearing impairment, peripheral neuropathy.
- Severe myelosuppression; dehydration.
- Pregnancy, breastfeeding.

### KEY INTERACTIONS

- **Aminoglycosides, loop diuretics, vancomycin, NSAIDs:** ↑ nephro/ototoxicity.
- **Phenytoin:** ↓ levels.
- **Live vaccines:** contraindicated.

## NURSING MANAGEMENT

- Pre-treatment: U&E, $Mg^{2+}$, CrCl, FBC, audiometry (baseline).
- Strict fluid balance, urine output monitoring, daily weight.
- Triple antiemetic prophylaxis (NK1 antagonist + 5-HT3 + dexamethasone) and delayed nausea plan.
- Replace $Mg^{2+}$/$K^+$; monitor for tinnitus, numbness.
- Cytotoxic handling and waste.

## PATIENT EDUCATION

- Drink plenty of fluids for several days after treatment.
- Take antiemetics regularly as prescribed, even if not sick.
- Report ringing in ears, hearing loss, numbness or tingling, reduced urine.
- Report fever ≥38 °C urgently.

## KEY POINTS

- Platinum DNA cross-linker.
- Nephrotoxic — hydration; oto- and neurotoxic.
- Highly emetogenic (acute and delayed).
- No aluminium needles.

**SOURCES CHECKED** **[874]** Cisplatin 1 mg/ml concentrate for solution for Infusion (UK SmPC) **[875]** Cisplatin, IV (US label) **[876]** Cisplatin, IV (US label, BluePoint Laboratories)
*Full references p. 432–457*